Kernix 0.12.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

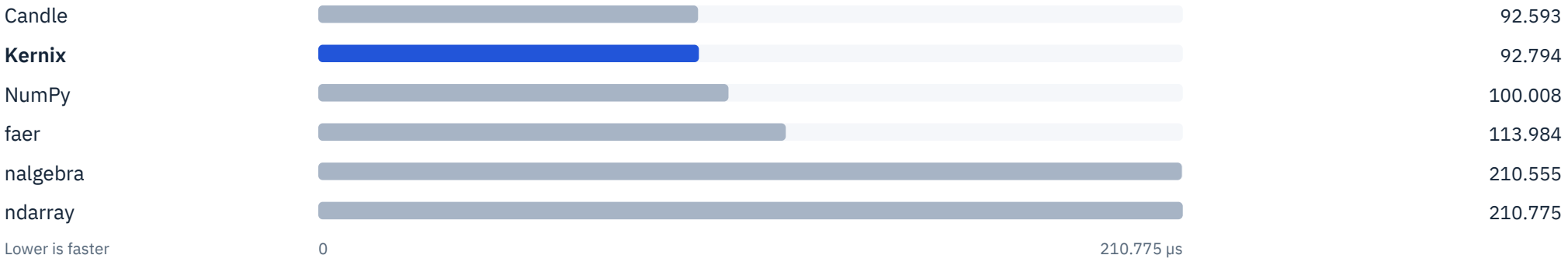
51 workloads

170 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.12.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.12.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



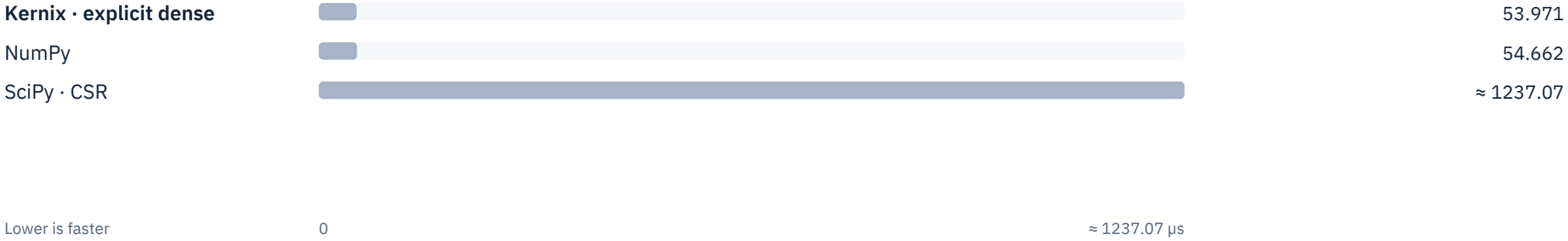
Kernix 0.12.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



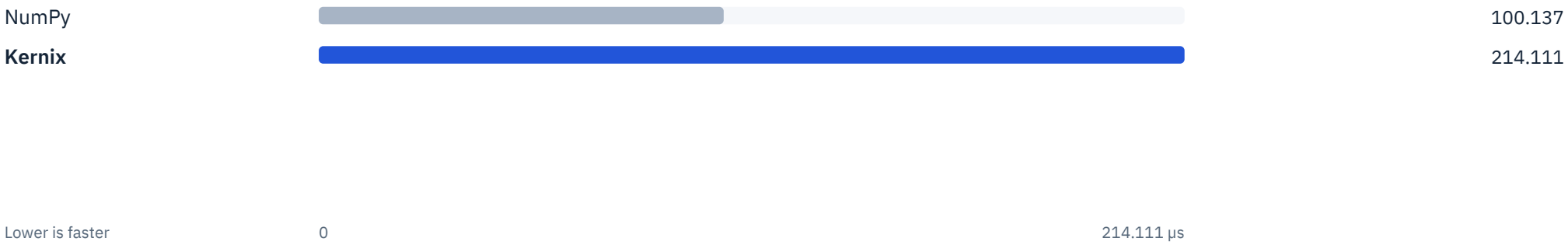
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

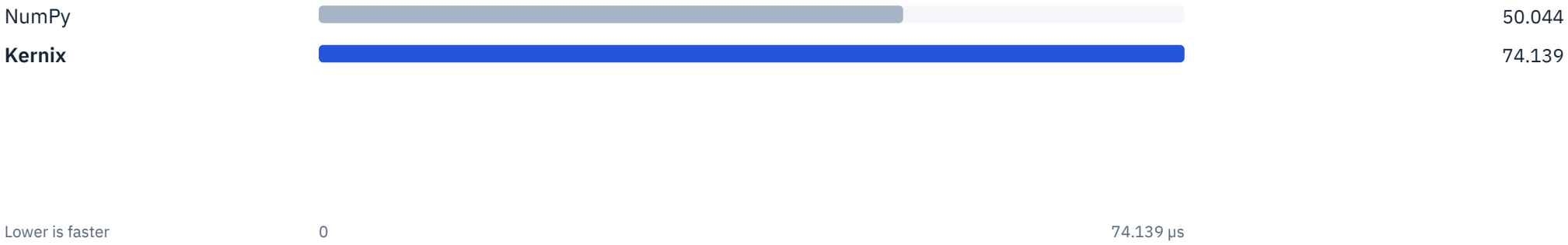
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



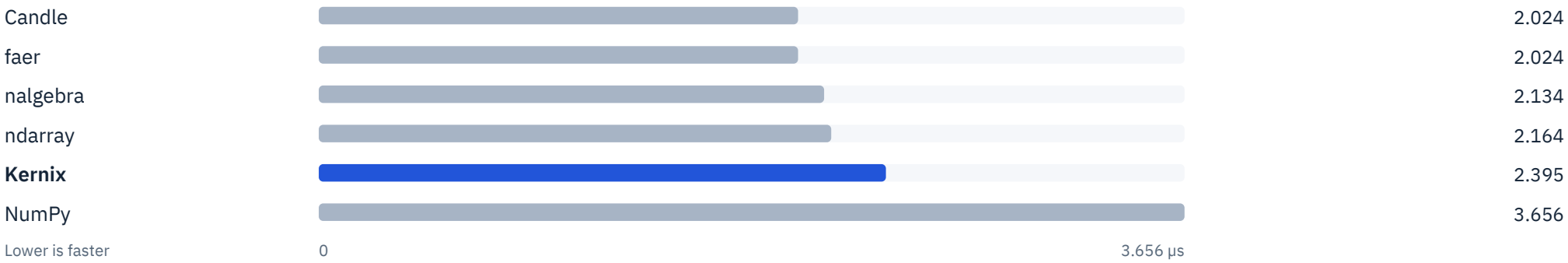
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

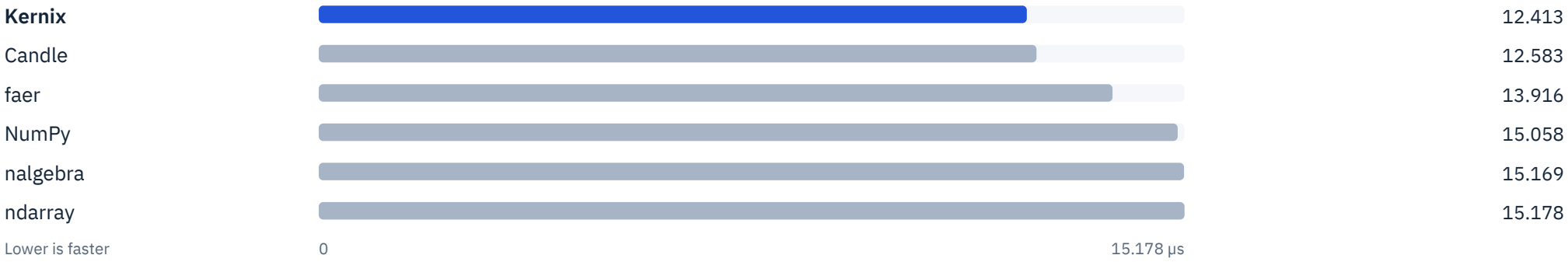
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



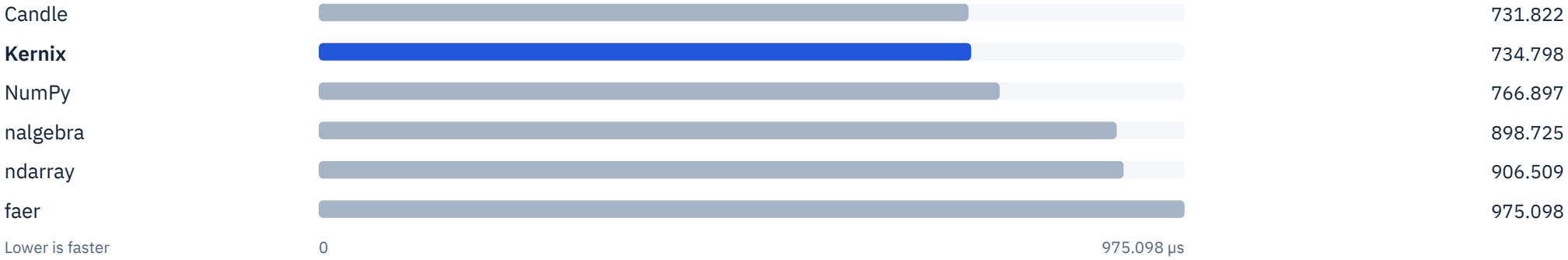
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

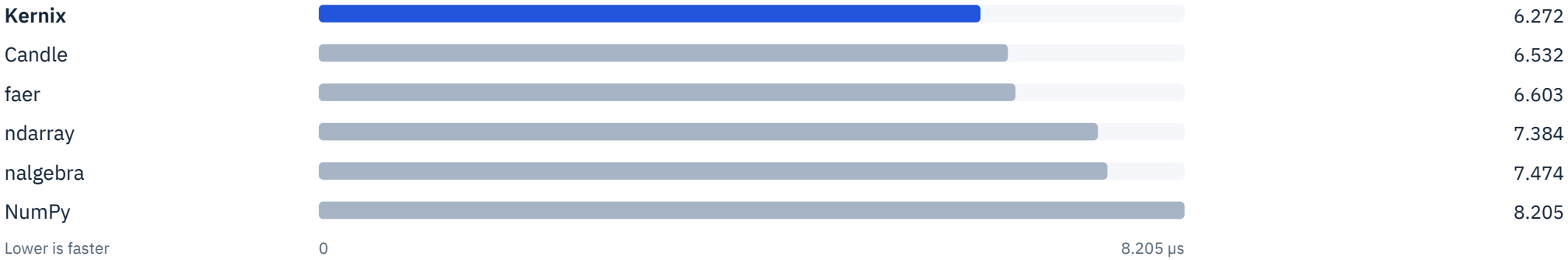
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



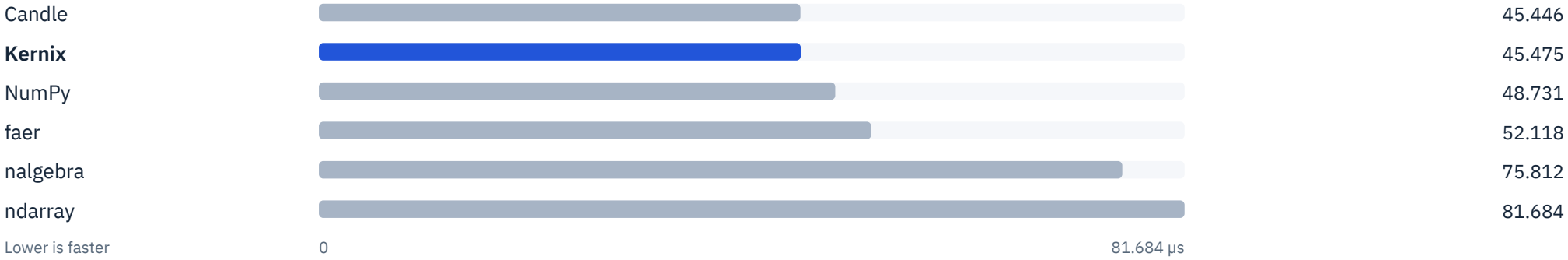
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

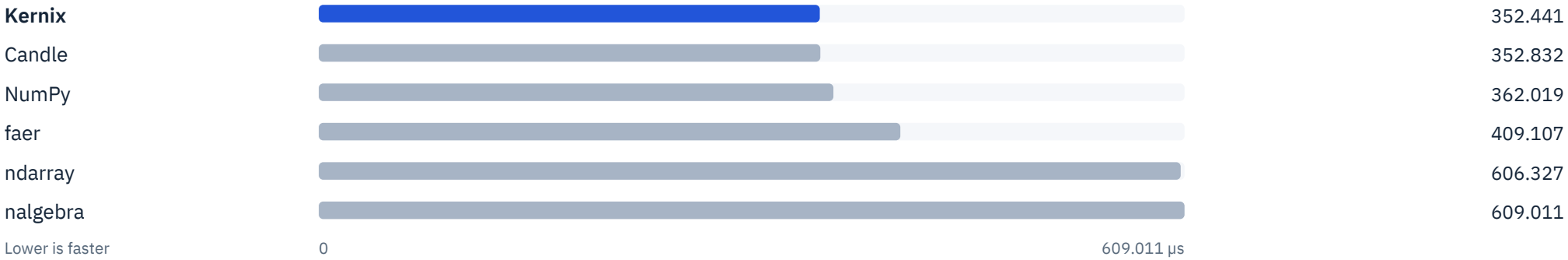
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



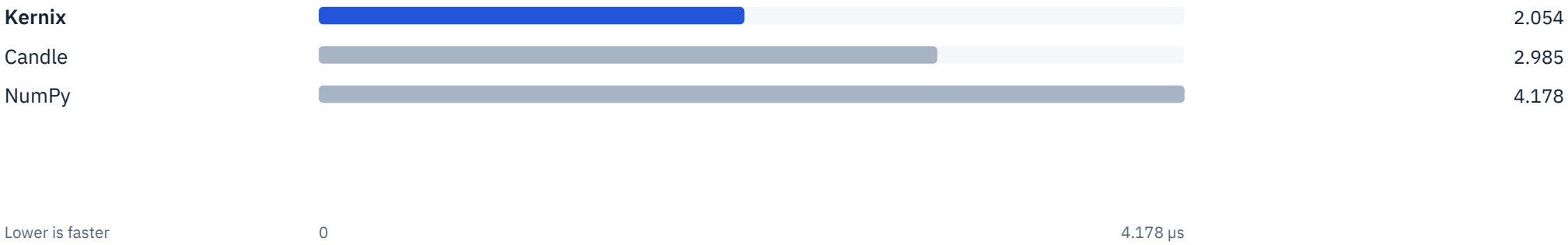
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

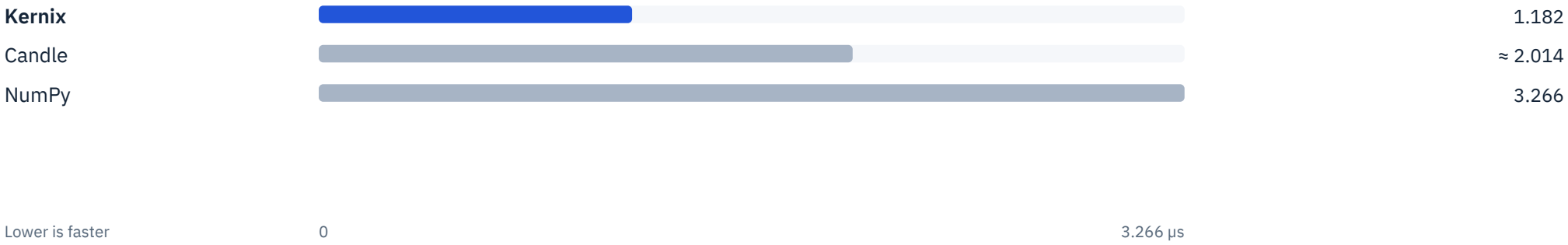
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



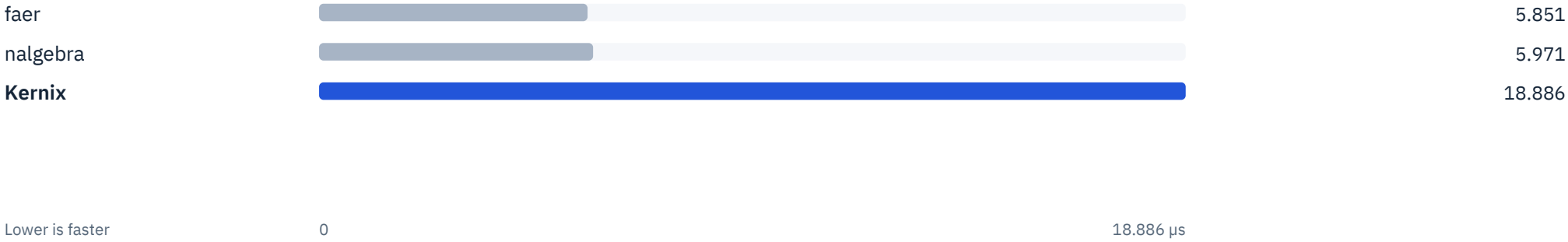
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



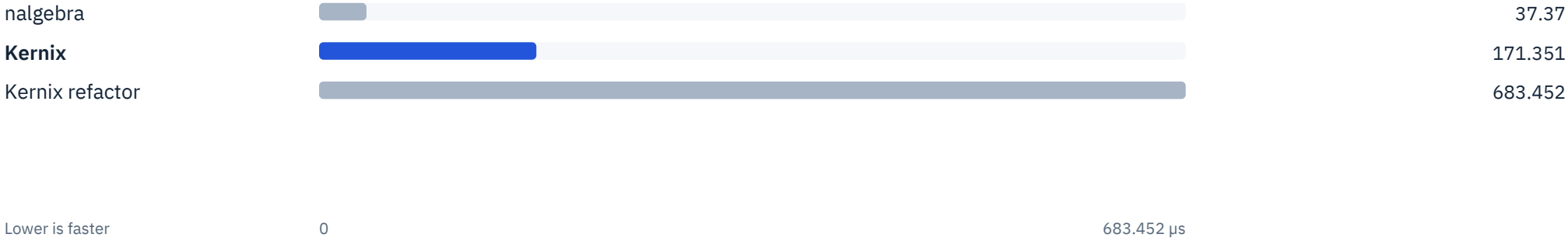
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

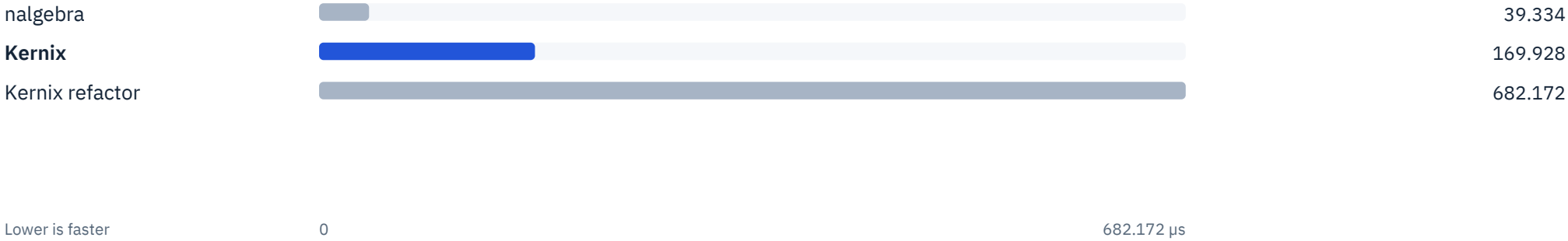
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

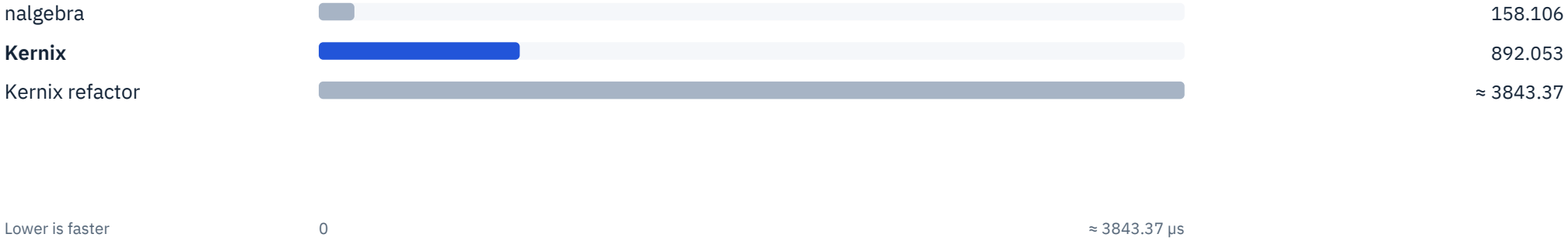


Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

512 × 128 · f64 · 1 worker · allocated output



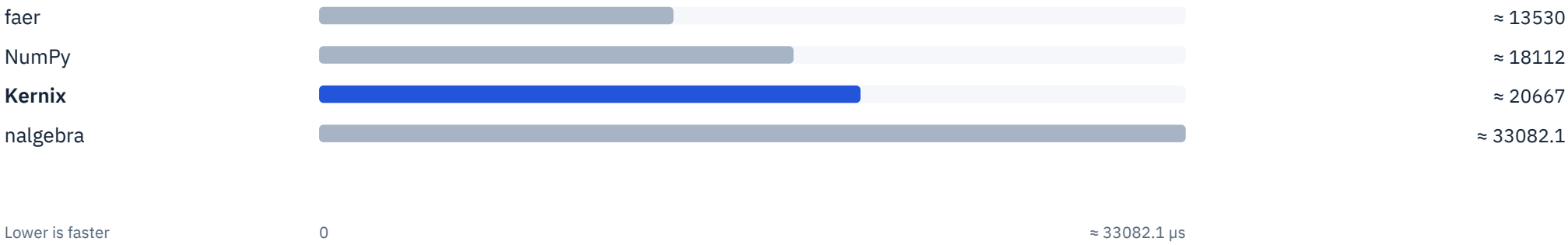
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

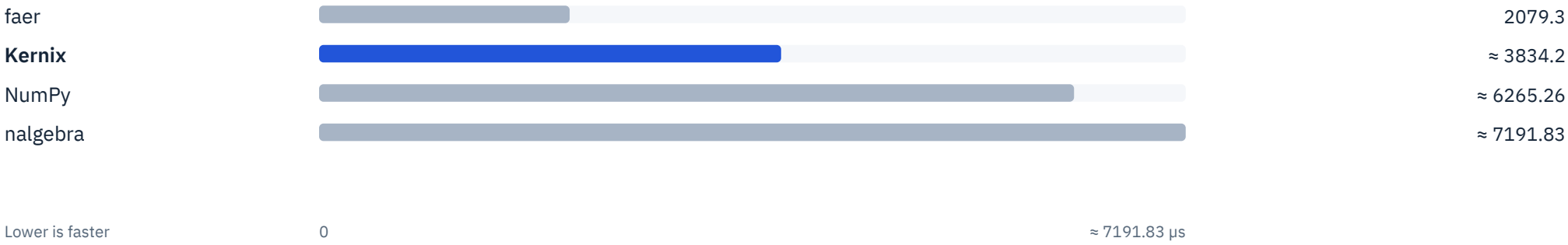
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



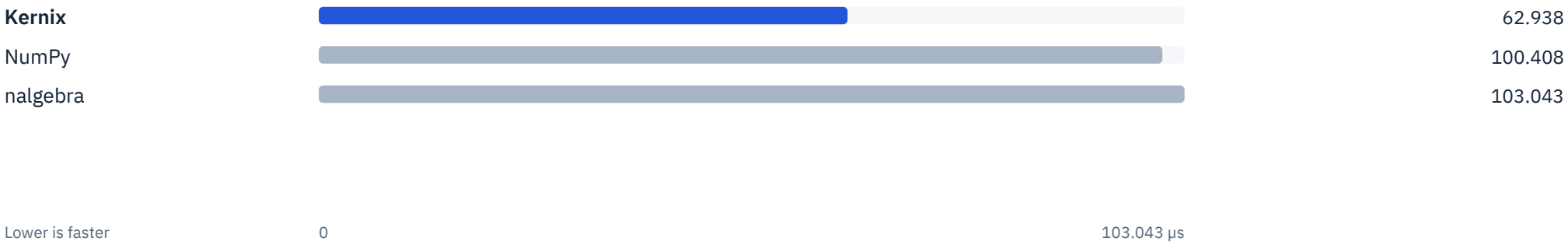
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

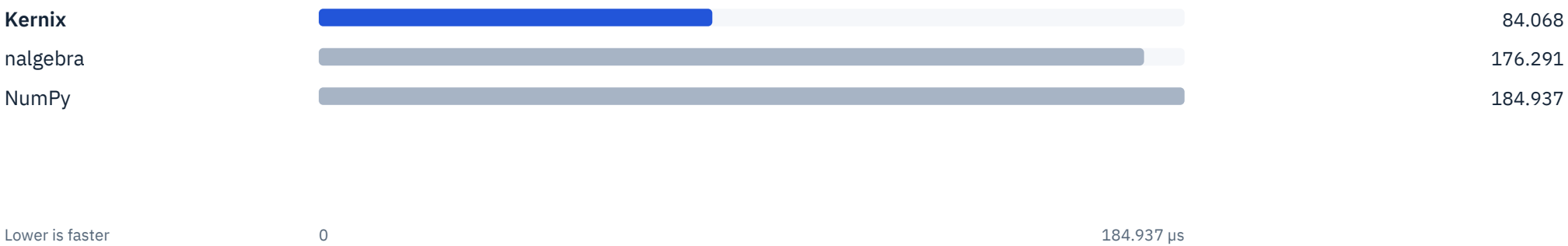
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



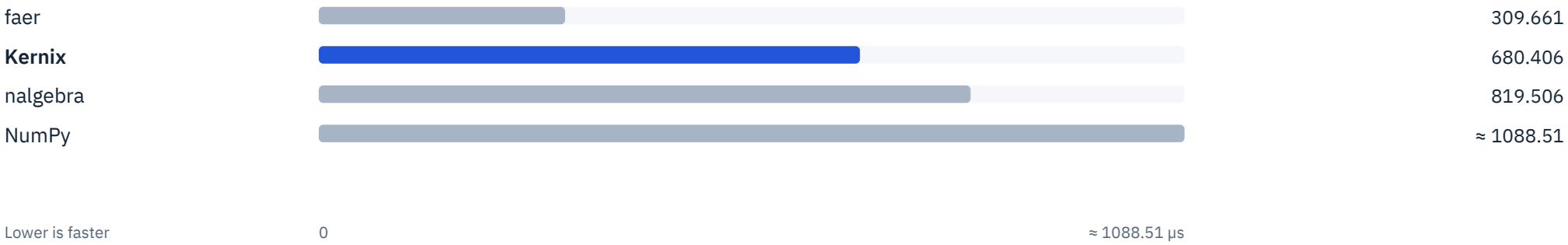
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

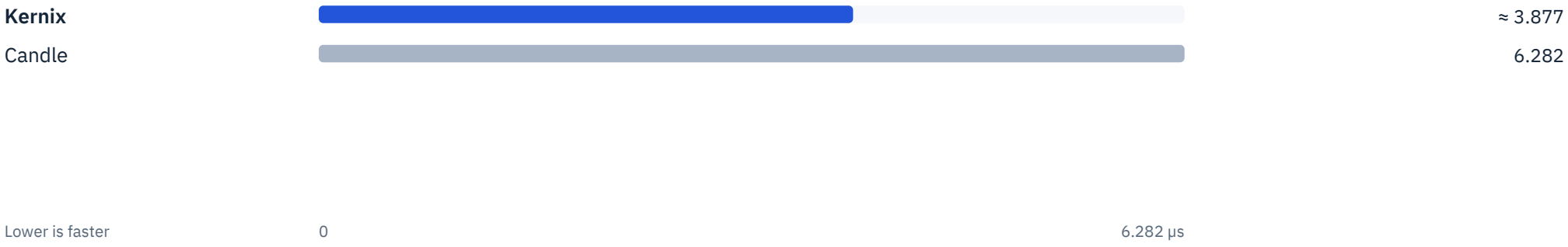
Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



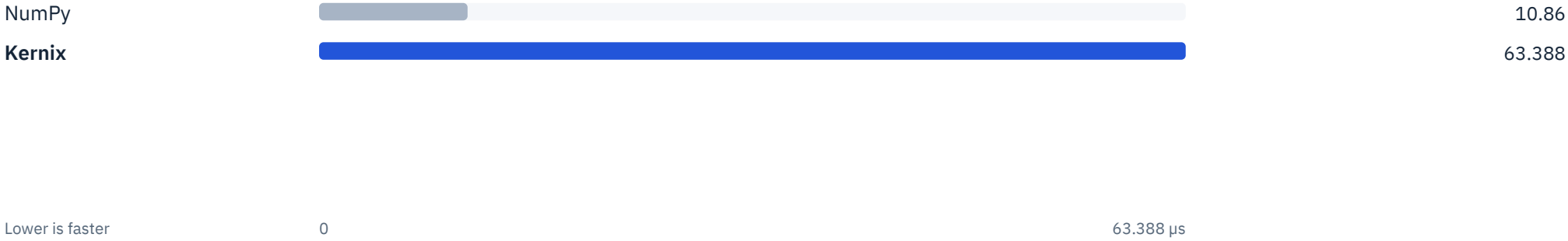
Kernix 0.12.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

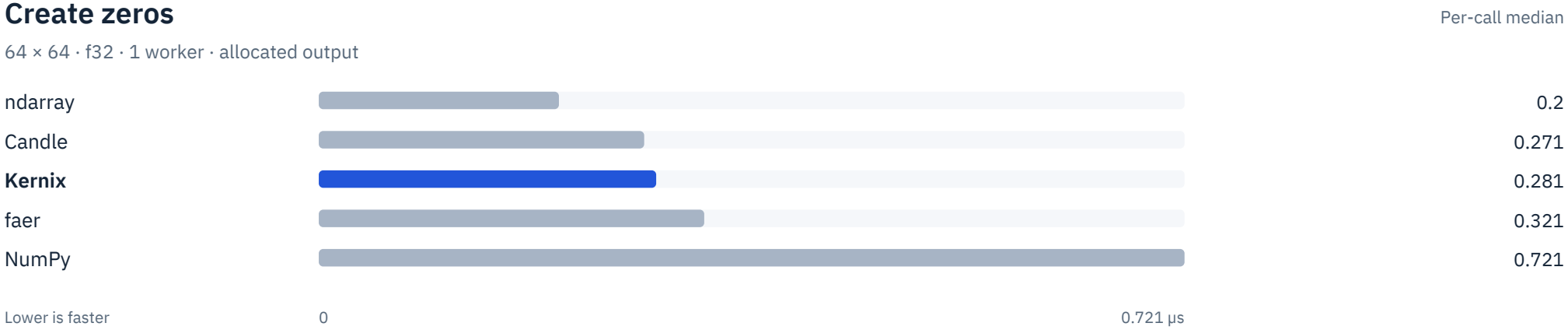


Kernix 0.12.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	10370	10.37
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	557595	557.595
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	7063	7.063
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	2838031	2838.031
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	26920	26.92
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	603211	603.211
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	474049	474.049
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	67146	67.146
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	18986	18.986
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	253525	253.525

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	61165	61.165
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	207649	207.649
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	65824	65.824
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	768180	768.18
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	731922	731.922
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	54662	54.662
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	1237069	1237.069
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	53971	53.971
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	100137	100.137
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	214111	214.111

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	74139	74.139
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	50044	50.044
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	100809	100.809
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	93004	93.004
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	45625	45.625
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	49233	49.233
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2024	2.024
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2134	2.134
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3656	3.656
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2164	2.164

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2024	2.024
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2395	2.395
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12583	12.583
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	15169	15.169
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15058	15.058
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12413	12.413
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13916	13.916
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15178	15.178
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	113984	113.984
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	210775	210.775

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100008	100.008
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	210555	210.555
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	92794	92.794
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	92593	92.593
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	766897	766.897
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	731822	731.822
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	898725	898.725
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	734798	734.798
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	906509	906.509
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	975098	975.098

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6272	6.272
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6532	6.532
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8205	8.205
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7474	7.474
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6603	6.603
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7384	7.384
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	45475	45.475
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	45446	45.446
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	48731	48.731
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	52118	52.118

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	75812	75.812
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	81684	81.684
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	409107	409.107
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	352441	352.441
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	606327	606.327
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	609011	609.011
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	362019	362.019
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	352832	352.832
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4178	4.178
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2985	2.985

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2054	2.054
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2014.0000000000002	2.0140000000000002
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1182	1.182
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	3266	3.266
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	15719.999999999998	15.719999999999998
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	56245	56.245
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	391394	391.394
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	30978	30.978
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	5971	5.971
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	18886	18.886

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	5851	5.851
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	275186	275.186
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1205710	1205.71
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1219647	1219.647
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	9558176	9558.176
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2205695	2205.695
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	9608824	9608.824
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1132363	1132.363
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	348844	348.844
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1182177	1182.177

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	683452	683.452
Cholesky rank-one update / nalgebra 256 × 256 · f64	37370	37.37
Cholesky rank-one update / Kernix 256 × 256 · f64	171351	171.351
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	682172	682.172
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	39334	39.334
Cholesky rank-one downdate / Kernix 256 × 256 · f64	169928	169.928
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	3843366	3843.366
Cholesky rank-one update / Kernix 512 × 512 · f64	892053	892.053
Cholesky rank-one update / nalgebra 512 × 512 · f64	158106	158.106
Thin QR / Kernix 512 × 128 · f64	3731610	3731.61

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 512 × 128 · f64	3420342	3420.342
Thin QR / faer 512 × 128 · f64	2251642	2251.642
Thin QR / nalgebra 512 × 128 · f64	4475523	4475.523
Thin QR / nalgebra 1024 × 256 · f64	33082110.999999996	33082.110999999996
Thin QR / Kernix 1024 × 256 · f64	20667047	20667.047
Thin QR / faer 1024 × 256 · f64	13530048	13530.048
Thin QR / NumPy 1024 × 256 · f64	18111968	18111.968
Cholesky factorization / faer 512 × 512 · f64	2079300	2079.3
Cholesky factorization / Kernix 512 × 512 · f64	3834198	3834.198
Cholesky factorization / nalgebra 512 × 512 · f64	7191831	7191.831

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 512 × 512 · f64	6265265	6265.265
Column-pivoted QR / faer 96 × 32 · f64	42830	42.83
Column-pivoted QR / Kernix 96 × 32 · f64	79068	79.068
Column-pivoted QR / nalgebra 96 × 32 · f64	76613	76.613
Column-pivoted QR / Kernix 256 × 64 · f64	792246	792.246
Column-pivoted QR / faer 256 × 64 · f64	381395	381.395
Column-pivoted QR / nalgebra 256 × 64 · f64	739035	739.035
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	62938	62.938
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	100408	100.408
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103043	103.043

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	184937	184.937
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	84068	84.068
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	176291	176.291
Thin QR / Kernix 96 × 32 · f64	66815	66.815
Thin QR / faer 96 × 32 · f64	55154	55.154
Thin QR / NumPy 96 × 32 · f64	68588	68.588
Thin QR / nalgebra 96 × 32 · f64	57639	57.639
Thin QR / nalgebra 256 × 64 · f64	556803	556.803
Thin QR / NumPy 256 × 64 · f64	542467	542.467
Thin QR / faer 256 × 64 · f64	418504	418.504

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 256 × 64 · f64	750187	750.187
Cholesky factorization / NumPy 64 × 64 · f64	17974	17.974
Cholesky factorization / Kernix 64 × 64 · f64	25036	25.036
Cholesky factorization / nalgebra 64 × 64 · f64	20468	20.468
Cholesky factorization / faer 64 × 64 · f64	14167	14.167
Cholesky factorization / nalgebra 128 × 128 · f64	120926	120.926
Cholesky factorization / NumPy 128 × 128 · f64	105979	105.979
Cholesky factorization / faer 128 × 128 · f64	57689	57.689
Cholesky factorization / Kernix 128 × 128 · f64	170329	170.329
Cholesky factorization / faer 256 × 256 · f64	309661	309.661

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	819506	819.506
Cholesky factorization / NumPy 256 × 256 · f64	1088511	1088.511
Cholesky factorization / Kernix 256 × 256 · f64	680406	680.406
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10310	10.31
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4568	4.568
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3647	3.647
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	5551	5.551
Model inference / Candle batched normalized/sliced MLP · f64	4348	4.348
Model inference / Kernix batched normalized/sliced MLP · f64	3026	3.026
Model recording + forward / Candle batched normalized/sliced MLP · f64	6282	6.282

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3877.0000000000005	3.8770000000000005
Model backward / Kernix prepared graph → fresh gradients · f64	4769	4.769
Model backward / Candle prepared graph → fresh gradients · f64	16831	16.831
Parameter update / Kernix transactional SGD, fixed gradients · f64	852	0.852
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63388	63.388
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	10860	10.86
Row softmax / Candle 256 × 64 · f64	84518	84.518
Row softmax / Kernix 256 × 64 · f64	94077	94.077
Row softmax / NumPy 256 × 64 · f64	119875	119.875
Row softmax / ndarray 256 × 64 · f64	80982	80.982

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	721	0.721
Create zeros / ndarray 64 × 64 · f32	200	0.2
Create zeros / faer 64 × 64 · f32	321	0.321
Create zeros / Candle 64 × 64 · f32	271	0.271
Create zeros / Kernix 64 × 64 · f32	281	0.281
Create ones / Candle 64 × 64 · f32	441	0.441
Create ones / faer 64 × 64 · f32	391	0.391
Create ones / ndarray 64 × 64 · f32	351	0.351
Create ones / Kernix 64 × 64 · f32	431	0.431
Create ones / NumPy 64 × 64 · f32	2043	2.043